TIM 80C, Lecture # 3 (4/7/26)

Agenda

0 : Complete Product dissection
done : see above

1. HW # 1, Prob. 1 : SPSP

2. HW # 2, Prob. 2 : Product Dissection

Comments :

1) Scope of the course : lower division, specifically aimed at 1st, 2nd year students in any Major

2) Your homework :
   - all text needs to type-written; not handwritten
   - diagrams can be hand-drawn & inserted in the text

1. HW # 1, Prob. # 1

Raw problem statement : "Improve the home computer."

Use the structured problem solving process (SPSP): customize the SPSP template on Canvas for this problem.

Step 1 : Define the real problem.

Assess existing home computers with respect to the user needs, & then develop guidelines & recommendations for improving the home-computer w.r.t. each of these needs.

Step 2 : Create a plan

(1) What are the different types of home-computers, & which type am I going to focus on for this problem

- laptop, desktop, server
- Dell, Apple, Lenovo, ...

⇒ focus on the computer type I'm familiar with.

(2) what are the typical user needs for home computers ?

(3) what are the key sub-systems of a home computer?
(Create a FAST diagram that relates user needs (sub-functions) to sub-systems.)

(4) Assess how well existing home computers satisfy user (or customer) needs.

(5) Create a table with guidelines & recommendations for improving home computers
(research, structured brainstorming)

Step 3 : Execute the plan

Rewrite each step of the above plan, and then for each step show the execution or implementation

: : : } execution
. . . .

⟹ Results

Table (of Results):

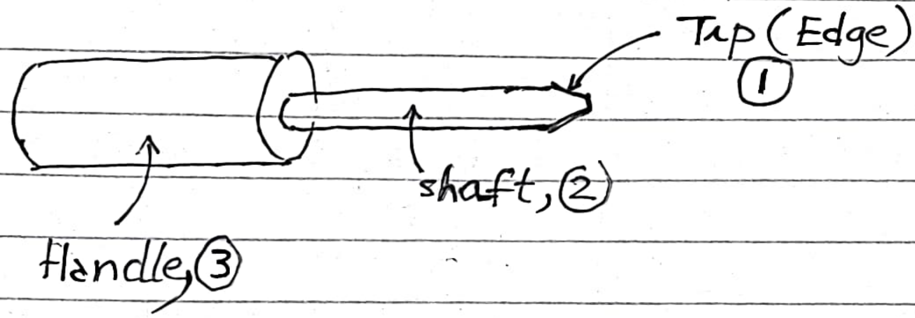
Customer needs (Step 2)	Assessment (Step 4)	Recommendations for improvement (Step 5)
Price		
Performance - speed - memory - ?		
Reliability - -		
- -		

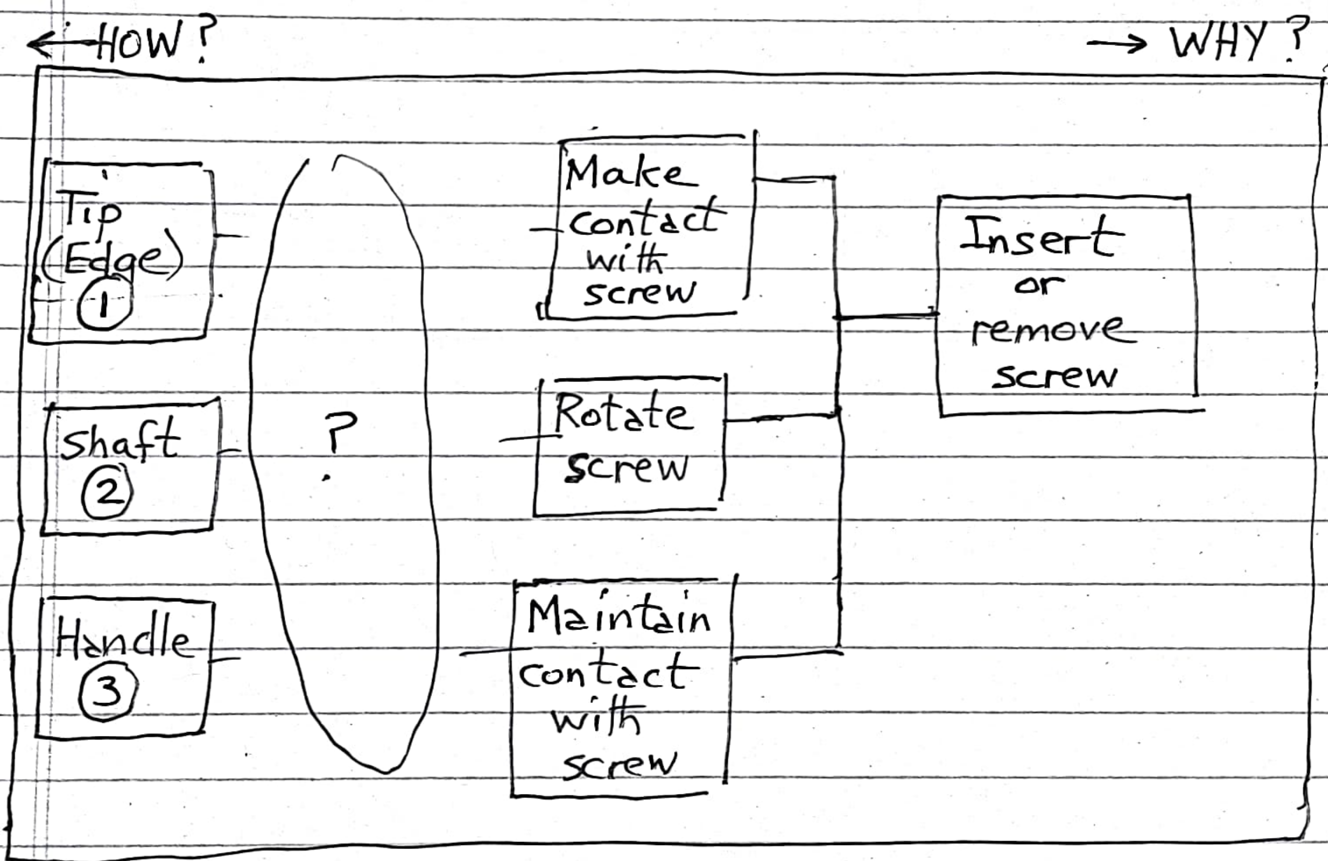
SPSP Step 4 : Check your work
- Are my results reasonable to me?
- Test your results on others

SPSP Step 5 : Learn ("Reflection")
- Learn & generalize from your results on the actual problem solved: draw conclusions
- your own problem-solving style.

2. HW # 1, Prob. 2 : Dissecting different products using FAST

Screwdriver :





Function Analysis System Technique (FAST) or Reverse Engineering or Product Dissection of complex products (e.g., bicycle)

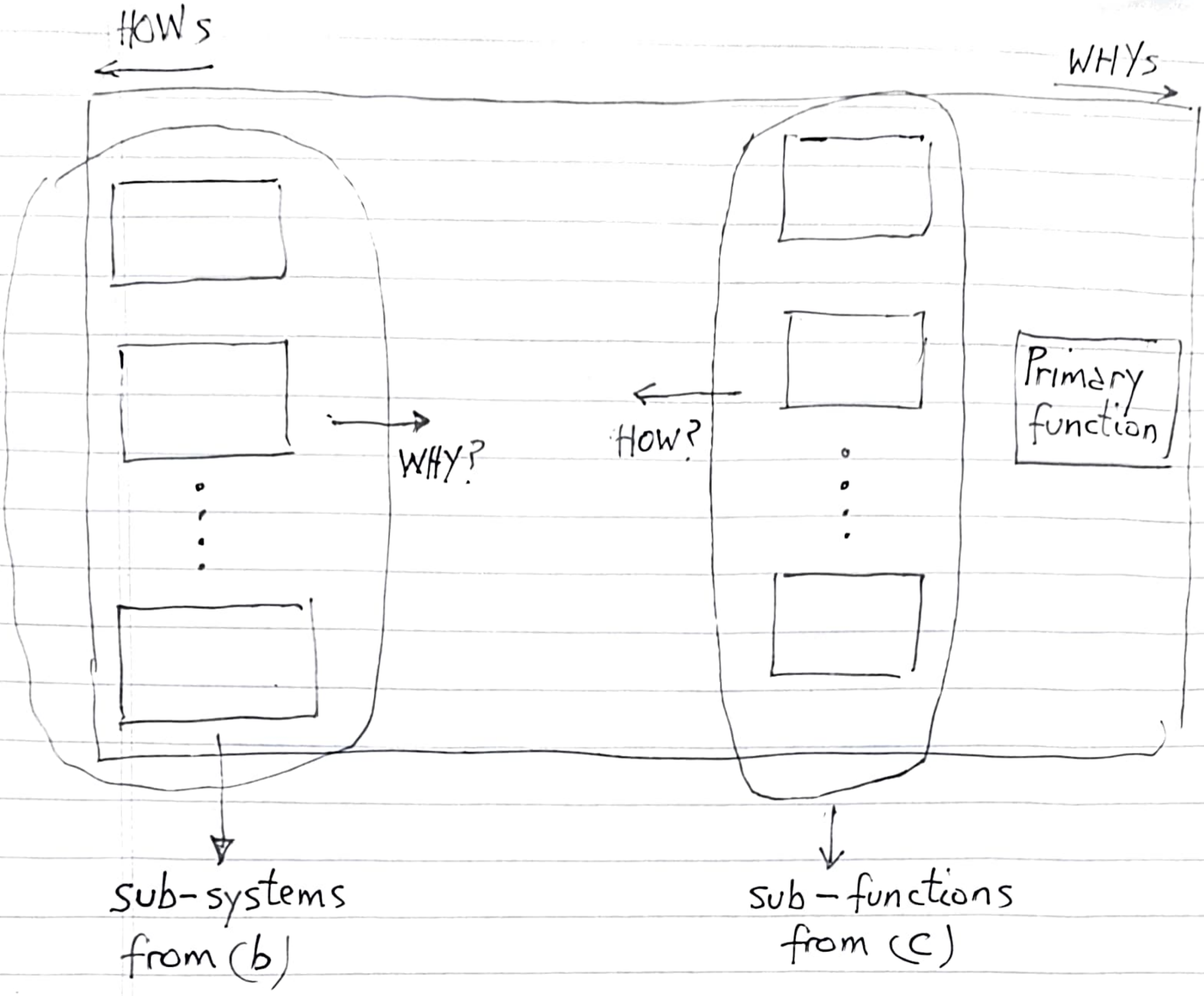
(a) Understand how the product works - (for example, howstuffworks.com)

(b) Make a list of the key sub-systems of the product.

(c) Identify the main function (or purpose) of the product, and make a list of the sub-functions of the product.

(d) Work from both ends of the FAST diagram:

$\longrightarrow$



<u>Observation</u> : This is an iterative, trial & error process ⇒ should converge in 2-3 iterations.

HW # 1, Prob. 2

Bicycle:

- understand how the bicycle works
  ⇒ make notes

Key sub-systems:

- Transmission
- Brakes
- Chassis (frame)
- Steering mechanism
- wheels
- pedals
- ....
  ....

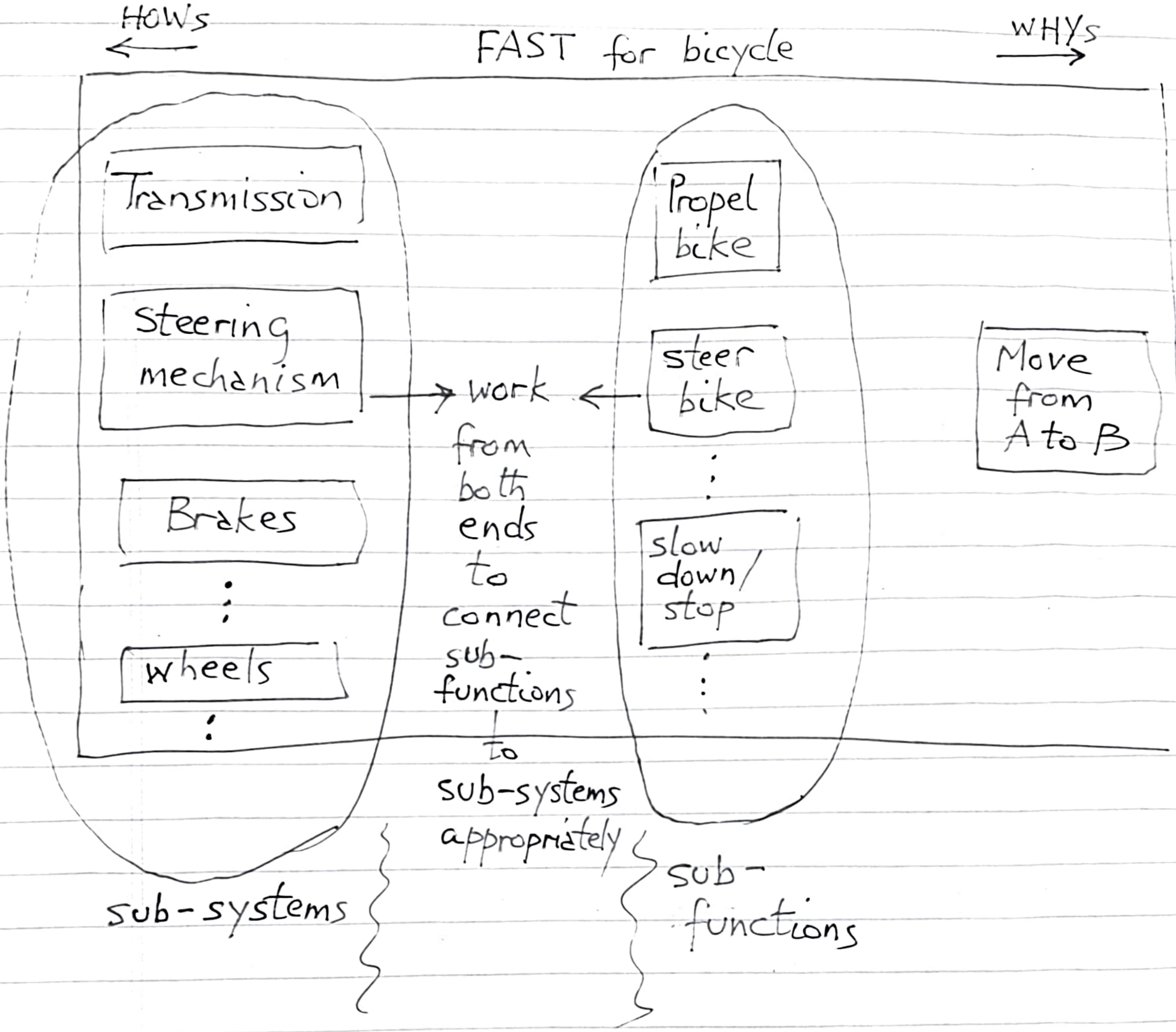
What are the key sub-functions

Primary function: Transport person from A to B

Sub-functions:

- Propel or move
- steer
- slow down/stop
- ...
- ...

FAST for bicycle



Laptop

Understand how the laptop works ⟹ make notes

Make a list of the key sub-systems

- Microprocessor (CPU)
- Operating System
- Keyboard
- Trackpad (Mouse)
- Screen
- Storage
- Battery
- ...
- ...

Make a list of the key sub-functions